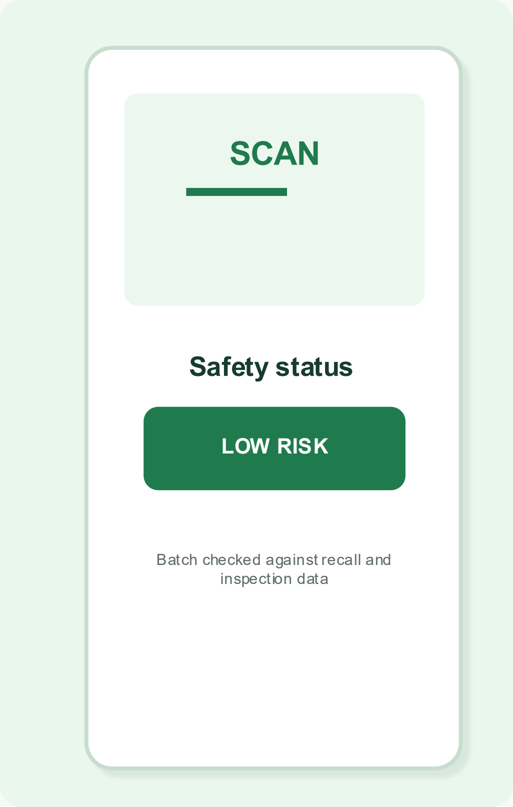
QR-Code Food Safety Inspector API

A portable information structure that turns fragmented food safety records into scan-ready consumer, supplier, and regulator insight.

Portable information product | Food safety + traceability | JSON API prototype



Jihan Yang, Raymond Xu| IMT542 Final Project

The information problem

Food safety information exists, but it is not portable at the decision point.

Today

Consumers scan labels, search recall pages, or trust incomplete packaging claims. Suppliers and regulators each hold pieces of the safety story.

Fragmented sources
Uneven formats
Slow reuse

Target state

A QR code resolves to a standardized API record that combines product identity, batch traceability, recall status, inspection context, and user-specific access views.

One scan
One portable record
Actionable safety signal

Info story and users

The same portable structure supports different access views.

Everyday consumer

Scan a food item before purchase or use.

Need: simple safety status, origin, recall warning.

Supply chain manager

Monitor batches, inventory risk, and source quality.

Need: batch history, supplier status, operational alerts.

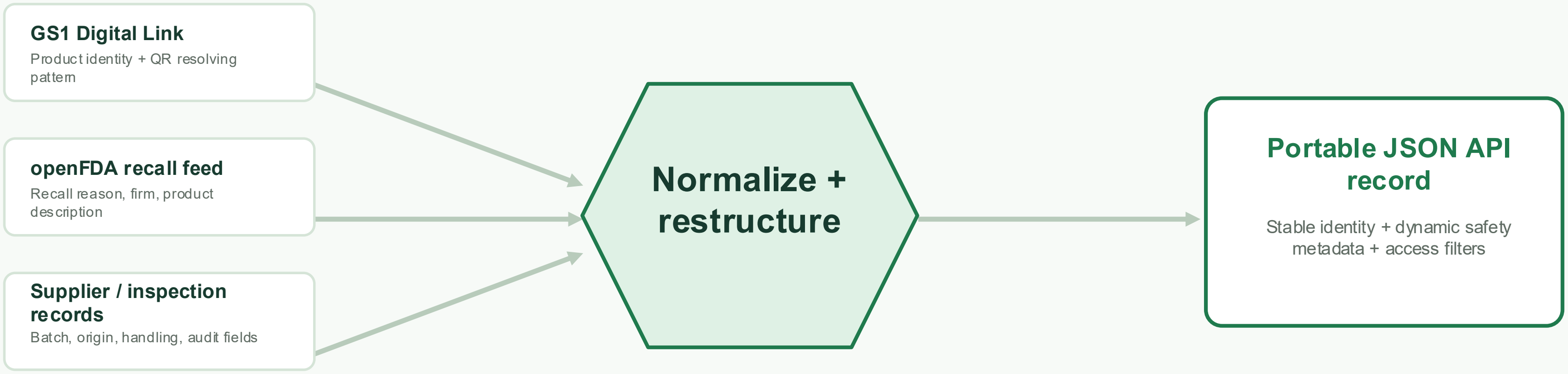
Health inspector

Verify records during audits or incident response.

Need: traceable evidence, update metadata, audit details.

Existing information structures

The prototype extracts and restructures existing information rather than inventing a new closed dataset.



Methodology: Extract fields from current sources, transform inconsistent names and values, then expose a reusable structure for web, mobile, CSV, and API use.

FAIR assessment of the current state

The current ecosystem is useful, but not yet reusable as one decision-ready safety record.

FAIR area	Rating	Why it matters for the information story
Findable	Medium	Product IDs and recall pages exist, but batch-level safety context is scattered.
Accessible	Mixed	Public recall data is accessible; supplier and inspection context often requires separate systems or permissions.
Interoperable	Weak-Medium	GS1 gives a strong identity pattern, but recall, inspection, and supplier fields use different schemas.
Reusable	Weak	Licensing, metadata, freshness, and field definitions are uneven across sources.

✓ **Improvement target**

Keep source traceability while standardizing field names, identifiers, timestamps, and access views.

✓ **Course focus**

Portability is improved through range, scope, frequency, and interoperability of information access.

Deficiencies that drive the redesign

The design choices come from the portability gaps found in prior assignments.

No single record

A consumer cannot easily see identity, recall status, origin, and handling context together.

Different update rhythms

Recall feeds, supplier data, and inspection records change at different frequencies.

Mixed audience needs

Consumer simplicity conflicts with regulator detail unless access views are separated.

Weak quality signals

A stale or incomplete record can look trustworthy unless freshness and validation are explicit.

New portable information structure

A hierarchical JSON schema separates stable identity from dynamic safety and access metadata.


```json
{
  "product_identity": { "gtin": "00012345678905", "name": "Organic Spinach" },
  "batch_identity": { "lot_number": "SP-2026-05-18", "pack_date": "2026-05-18" },
  "safety_and_quality_information": { "temperature_status": "pass", "inspection_score": 96 },
  "recall_information": { "recall_status": "none", "last_recall_check": "2026-05-29" },
  "supply_chain_information": { "origin_farm": "WA-FARM-102", "processing_site": "WA-PROC-14" },
  "consumer_access_view": { "risk_level": "low", "message": "No active recall found" },
  "regulator_audit_metadata": { "source_records": ["openFDA", "supplier_db"] }
}
```


✓ Stable identity

GTIN, product name, lot, and pack date stay consistent across users and tools.

✓ Dynamic safety metadata

Recall status and quality checks can update without changing the product identity.

✓ Role-based views

Consumer view is simple; supplier and regulator views keep deeper operational evidence.

✓ Reusable formats

The same structure can be exported as JSON, CSV, or served through REST endpoints.

Prototype access flow

The QR scan turns a physical food item into an API-backed information product.

1. **QR on package** — GS1 Digital Link / product URL
2. **API request** — GET batch safety record
3. **ETL + validation** — Merge recall + supplier + inspection
4. **Access filter** — Consumer / supplier / regulator view
5. **Actionable output** — Risk level + details + next action

Design result: the QR code is not just a label. It becomes the access point for a portable, versioned, and testable information structure.

Quality, performance, and security controls

The test plan makes the prototype a living information system, not a one-time demo.

Functional tests

- QR resolves to correct endpoint
- Schema validation for every JSON record
- Recall sync updates recall_status
- Consumer view hides backend audit fields

Performance tests

- p95 API response target under 1 second for cached records
- Recall refresh job monitored for completion
- CSV / JSON export checked for row and field completeness

Alarms + actions

- 5xx errors, stale last_recall_check, failed ETL job
- Unauthorized update attempts
- Action: retry feed, rollback bad data, notify owner, document incident

Outcome and GitHub deliverables

Community outcome

Consumers gain faster safety insight; suppliers gain reusable batch records; regulators gain more traceable and standardized audit context.

Repository structure

- README.md - project story, methods, setup
- /data - raw and transformed sample files
- /src - ETL and API prototype code
- /docs - FAIR, portability, deficiencies, test plan
- /presentation - final PPTX

Main takeaway

Portability improves when food safety information uses stable IDs, interoperable JSON fields, access views, and quality controls.

Questions